

A photograph of a large airport terminal at dusk. The terminal has a curved, glass-fronted roof and is illuminated from within. In the foreground, a wet tarmac reflects the lights of the terminal and the sky. Several ground service vehicles and a large white aircraft with a yellow tail are visible. An airplane is flying in the sky above the terminal.

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How Can Fuel Price Affect the Fuel Burn of Different Aircrafts?

Donghang Li, ChenAn Shen, Yuhan Tang

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Abstract

Fuel prices constitute a major operational expense for airlines, and price fluctuations can profoundly impact their fuel management strategies. This study analyzes historical domestic flight data to examine how variations in fuel prices affect fuel consumption across different aircraft types. Using interpretable machine learning with a Random Forest model and Shapley Value analysis, the findings reveal that fuel consumption is highly predictable, achieving an R^2 of 0.981, with fuel prices exerting a negative influence on consumption. Notably, fuel prices have a significant lagging effect, with the strongest impact observed two quarters prior. Additionally, wide-body aircraft generally exhibit better fuel efficiency adaptation to price increases, suggesting greater potential for efficiency improvements in larger aircraft. In contrast, the Boeing 737 series demonstrates a lower responsiveness to rising fuel prices. These insights are critical for optimizing fuel management strategies, especially as airlines navigate the financial challenges of transitioning to sustainable aviation fuel (SAF).

Introduction

The global aviation sector is committed to achieving net-zero emissions by 2050, with sustainable aviation fuels (SAFs) playing a pivotal role in reducing life-cycle carbon emissions by at least 10% compared to conventional jet fuel. SAFs have the potential to mitigate up to 65% of the projected 1.8 billion tonnes of carbon dioxide emissions by 2050, positioning them as a critical pathway for aviation decarbonization. However, widespread adoption faces significant barriers, including high costs—currently 2 to 4 times that of conventional jet fuel—and limited scalability. Given that fuel expenses account for a substantial share of airline operational costs, these price disparities may drive airlines to implement cost-control measures, particularly in non-en-route fuel management, encompassing taxi fuel, alternate fuel, tankering fuel, and discretionary fuel. Notably, unnecessary fuel loading, which contributes to 4.48% of average flight fuel consumption, costs airlines an estimated \$223 million annually. This study examines how airlines adjust fuel loading strategies in response to fuel price fluctuations, focusing on the elasticity of fuel loading and potential operational lags, using interpretable machine learning and multi-linear regression to analyze these dynamics.

Results

1.Exploratory Data Analysis (EDA)

Our analysis focuses on the fuel consumption of domestic flights for eight aircraft types across 29 airlines. We include only aircraft types with at least 80 observations (i.e., quarters of operation). We also exclude airline-aircraft pairs with fewer than two appearances to ensure data consistency. This filtering results in a dataset with 1,967 observations, each characterized by the airline, aircraft type, and time period (in quarters).

Figure 1(a) illustrates the total fuel consumed by the selected eight aircraft types over the 10-year period. Narrow-body aircraft show higher fuel consumption due to their predominance in domestic operations. Newer-generation aircraft, such as the Boeing 737 Next Generation (NG), which includes both B737-800 and B737-700, and the Boeing 777-300, which includes regular and extended-range versions, show increasing popularity. Conversely, older-generation aircraft, such as the Boeing 757-200, exhibit a decline in usage. A notable anomaly is the Boeing 737-900, which shows a sharp decline in fuel consumption in 2018. This can be attributed to

reporting changes: until the second quarter of 2018, United Airlines reported both B737-900 and B737-900ER operations under the B737-900 label, whereas Alaska Airlines reported these separately.

Figure 1(b) shows fuel consumption by major passenger airlines over the years. The decade under study includes several major airline mergers, resulting in the termination of reports from Continental Airlines, US Airways, and Virgin America. Consequently, we observe increases in fuel consumption for United Airlines, American Airlines, and Alaska Airlines post-merger. The slight decrease in fuel consumption immediately after these mergers likely reflects fleet consolidation and the retirement of older aircraft.

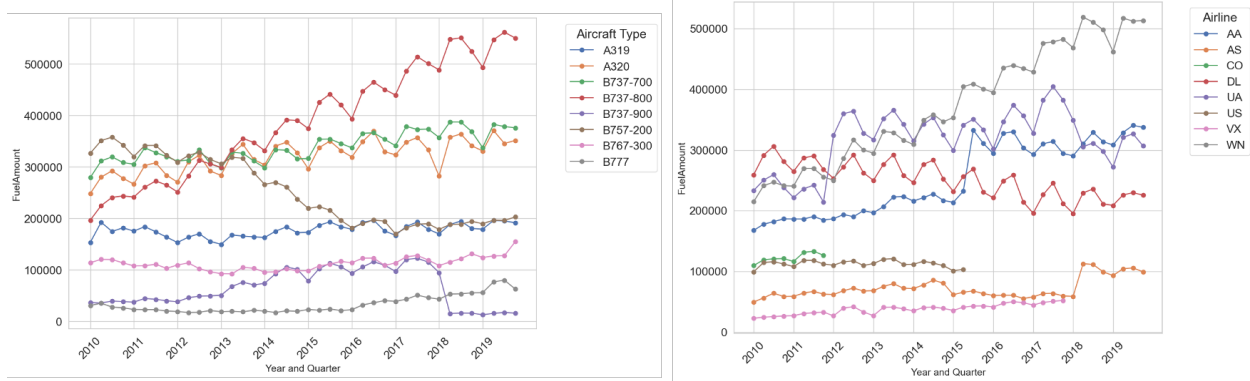


Figure 1(a). Fuel loading for different aircraft types over time

Figure 1(b). Fuel loading for a selected sample of airlines over time

Figure 2(a) shows fuel prices before and after inflation adjustment, revealing that the adjusted fuel price is lower than the nominal price. The price initially experienced relatively mild fluctuations, hitting its lowest point in 2016, followed by a gradual recovery.

To identify significant relationships between numerical variables, we analyzed their correlations by creating a correlation matrix and visualizing it with a heatmap, as is shown in Figure 2(b). The heatmap revealed several notable observations. For example, *Payload* shows strong correlations with *Distance*, *AirborneHours*, and *FuelAmount*, indicating that larger payloads are associated with longer flights and higher fuel usage. Additionally, *Distance* and *AirborneHours* are highly correlated, reflecting that longer distances naturally involve more air time. *FuelAmount* has strong positive correlations with *Payload*, *Distance*, and *AirborneHours* suggesting that flights using more fuel tend to be longer and carry more payload.

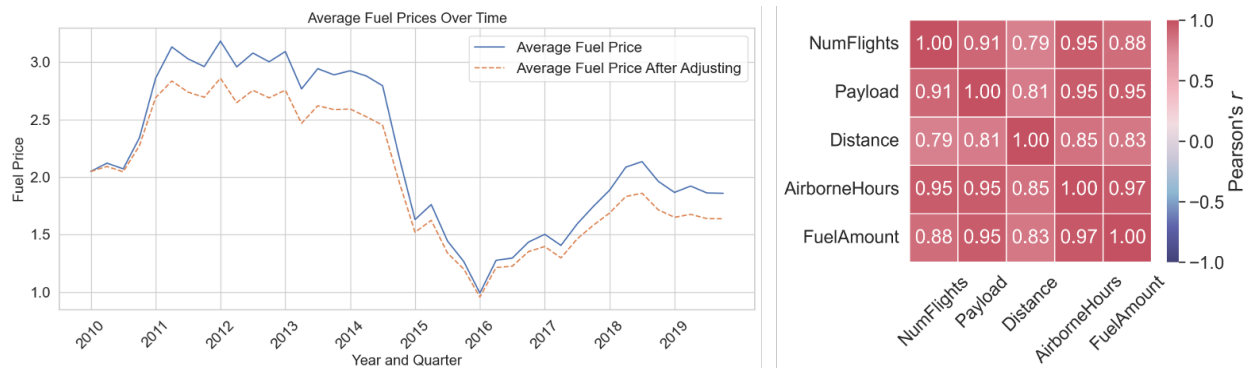


Figure 2(a). Average Jet-A fuel price over time

Figure 2(b). The correlation matrix between different variables

2. Interpretable Machine Learning Results

We employed four machine learning models to predict aviation fuel consumption, including Linear Regression, CART, Random Forest, and XGBoost. Table 1 presents the performance of these models. As shown, the Random Forest model achieved the best performance, with an out-of-sample R^2 of 0.981 and an RMSE of only 0.256. Therefore, we selected the Random Forest model as our final selected model.

Table 1. Machine Learning Models Performance

Methods	Training		Testing	
	R2	RMSE	R2	RMSE
Linear Regression	0.982	0.245	0.979	0.263
CART	1.000	0.000	0.964	0.344
Random Forest	0.997	0.092	0.981	0.256
XGBoost	0.999	0.050	0.978	0.271

To enhance the interpretability of the Random Forest model, we introduce the concept of the Shapley Value, which is a game-theoretic approach used to explain the contribution of each feature in a machine learning model. It calculates the average marginal contribution of a feature by considering all possible combinations of features, ensuring a fair attribution of the prediction.

Figure 3 illustrates the SHAP analysis results. The left panel shows a SHAP summary plot, where dots represent feature impacts on predictions. A blue-to-red color gradient indicates feature values from low to high. Notably, features like *ln_airborne_hours* and *ln_payload* show higher SHAP values for larger feature values, indicating a positive effect on the model output. The right panel contains individual dependence plots, showing how SHAP values change with feature values, with orange lines representing fitted trends.

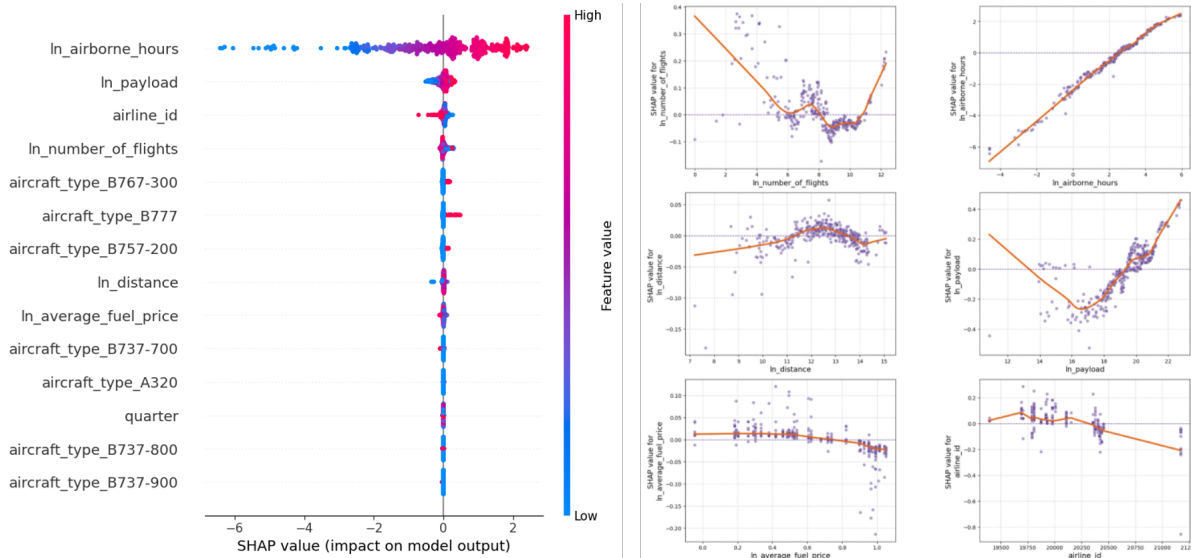


Figure 3. Shapley value results using Random Forest algorithm

Particularly, the feature *ln_average_fuel_price* has a relatively small and mostly negative impact, suggesting that higher fuel prices may be associated with reduced fuel consumption, potentially due to efficiency measures or operational adjustments by airlines. This result highlights the significance of fuel price fluctuations in aviation fuel consumption.

3. Heterogeneity Analysis: Disparities across Time and Aircraft Types

Fuel prices have a significant negative effect on aviation fuel consumption, and we want to see if the impact exhibits a dynamic pattern over time. Figure 4(a) shows the time variation results. We found that initially, the effect decreases but later intensifies, reaching its peak influence with a two-period lag. The fuel price with a two-period lag has the most significant impact on fuel consumption, which highlights the delayed response of fuel consumption to price changes, emphasizing the importance of considering temporal dynamics in fuel management strategies.

Figure 4(b) illustrates the effects across different aircraft types. Wide-body aircraft generally demonstrate better fuel efficiency adaptation to price increases, indicating greater potential for mitigating rising fuel costs. Among specific models, the Boeing 737 series exhibits a relatively limited response to fuel price increases, while the Airbus 319 stands out with excellent fuel efficiency adaptation, reflecting its strong potential for fuel cost management. This underscores the importance of considering the impact of different aircraft types when managing fuel consumption.

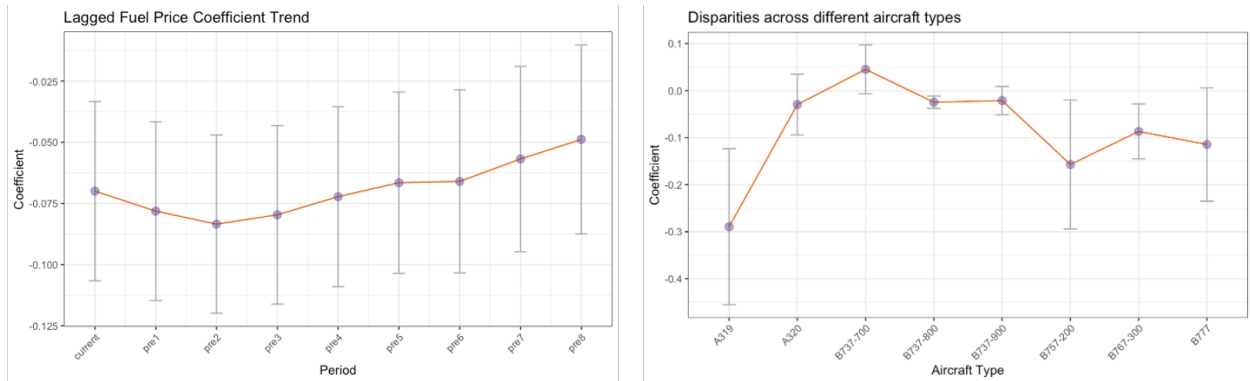


Figure 4(a). Time Variant Effect

Figure 4(b). Heterogeneity Effect across Aircraft Types

Conclusion

This study demonstrates that aircraft fuel consumption is highly predictable, with the Random Forest model achieving excellent predictive performance ($R^2 = 0.981$). Fuel prices have a significant negative effect on fuel consumption, influenced by both time and aircraft types, with the strongest impact observed at a two-period lag. Wide-body aircraft generally exhibit better fuel efficiency adaptation to fuel price changes, highlighting their potential for operational cost management. These findings are critical for optimizing fuel strategies, particularly as airlines face the financial and operational challenges of transitioning to sustainable aviation fuel (SAF). Incorporating predictive models and aircraft-specific strategies can support more effective and adaptive fuel management practices in the evolving aviation industry.

Appendix

1. Data Sources

To construct the model, we obtain three datasets, which characterize the volume of flight operations, airline fuel loading, and fuel price, respectively. Data on flight operations and fuel loading are collected from Form 41 - Air Carrier Financial Reports, which encompass a myriad of financial and operating information reported by U.S. airlines to the U.S. Department of Transportation (DOT). While these financial reports submitted to the DOT are generally unaudited, they must align with the airlines' GAAP reporting. Data spanning from 1990 to 2023 is available. Our research focuses on data from domestic flights in the U.S. between 2010 and 2019, excluding the effects of COVID-19 and the 2008 financial crisis.

From Form 41, we use two key datasets: Schedule P-5.2 and T-100 Domestic Segment (U.S. Carriers). While Schedule P-5.2 provides data on the total amount of fuel issued, T-100 records the volume of flight operations. The variable *FuelLoading* represents the amount of fuel loaded on aircraft. *AirborneHours* captures the number of hours aircraft are in the air, specifically between wheel-off and wheel-on time (excluding taxiing time). Additionally, *Distance* refers to the scheduled distance between the departure and arrival airports, not the actual miles flown.

We obtain historical fuel prices from the U.S. Energy Information Administration, which reports the free-on-board price for conventional Jet-A fuel. To account for inflation, we use the Producer Price Index (PPI) from 2010 to 2019. The PPI data, released monthly, is averaged over three months to provide a quarterly value. Similarly, we compute the quarterly average fuel price. The PPI data used is not seasonally adjusted. The discount factor for time period t is calculated as:

$$\pi_t = PPI_t / PPI_0$$

where π_t is the discount factor for time period t , PPI_t is the PPI for quarter t , and $PPI_0 = 173$ is the average PPI in the first quarter of 2010. The inflation-adjusted fuel price p^*_t is then computed as:

$$p^*_t = p_t / \pi_t$$

This transformation standardizes fuel prices to 2010 dollars, removing the effects of inflation for the subsequent machine learning models.

2. Data Structure

The data used in this study are organized csv data, and the descriptions of each variable in the data are shown in Table 2. Among all the variables, *airline_id*, *name*, *aircraft_type*, and *quarter* are factor variables. All the other variables are logarithm numbers. From *ln_fuel_price_t_minus_1* to *ln_fuel_price_t_minus_8* are a series of lagged variables of fuel prices in different quarters. In the dataset, we have 18 variables and 1,967 observations in total.

Table 2. Data Descriptions

Variables	Description	Format
airline_id	The id number of different airline companies	factor
name	The name of different airline companies	factor
aircraft_type	The type of aircrafts	factor
ln_fuel	The logarithm of fuel consumption	num
ln_number_of_flights	The logarithm of number of flights performed	num
ln_airborne_hours	The logarithm of total amount of time that an aircraft spends in the air	num

ln_distance	The logarithm of scheduled distance between the departure and arrival airports	num
ln_payload	The logarithm of available payload capacity	num
ln_average_fuel_price	The logarithm of fuel price current quarter	num
ln_fuel_price_t_minus_1	The logarithm of fuel price one quarter ago	num
ln_fuel_price_t_minus_2	The logarithm of fuel price two quarters ago	num
ln_fuel_price_t_minus_3	The logarithm of fuel price three quarters ago	num
ln_fuel_price_t_minus_4	The logarithm of fuel price four quarters ago	num
ln_fuel_price_t_minus_5	The logarithm of fuel price five quarters ago	num
ln_fuel_price_t_minus_6	The logarithm of fuel price six quarters ago	num
ln_fuel_price_t_minus_7	The logarithm of fuel price seven quarters ago	num
ln_fuel_price_t_minus_8	The logarithm of fuel price eight quarters ago	num
quarter	The quarter of flight	factor

3. Methods

Although the Random Forest regression model can identify key factors influencing aviation fuel consumption, it does not provide detailed insights into how each feature individually contributes to the predictions. To enhance model interpretability, we apply SHAP analysis, a model-agnostic approach recognized for its consistency and effectiveness, offering an advantage over other interpretative methods such as LIME. SHAP values quantify the influence of each independent variable, clarifying their specific contributions to fuel consumption predictions.

The contribution of each feature, denoted as ϕ_i , is calculated using the following formula:

$$\phi_i = \sum_{S \subseteq N} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

where $|S|$ is the number of features in subset S , n is the total number of features in set N , $v(S)$ represents the model output with only the features in subset S , and $v(S \cup \{i\})$ is the output when feature i is added to the subset. This approach considers all possible combinations of features, providing a comprehensive view of each feature's importance in predicting aviation fuel consumption within the Random Forest model.